Chapter 11 | Web Technologies (HTML / CSS / XML / JS / ASP)

11.1 The Web — Internet, Browsers, URLs and Site Design

(1) The internet is a network of networks (asked twice — also as 'a large network of networks'); it was **initially used for military intelligence** (ARPANET). The **WWW's primary function is to connect and access documents and resources over the internet**, and browsing websites and online services is its typical application. The **ISP is a gateway of the internet**.

(2) Browsers vs search engines: a **web browser displays and navigates web pages**; a **search engine is the specialised program that locates information on the Web** (Google, Yahoo, Lycos — **Internet Explorer is a browser, NOT a search engine**). Non-browsers planted in options: **MINIX (an OS)** and **Acrobat Reader (a PDF viewer)**; Mosaic, Safari, Opera, Edge, Chrome, Firefox are genuine browsers. Browser **developer tools inspect and debug web pages**; a browser loads a page of many small images faster **by having multiple threads** (parallel downloads). **location.reload( ) refreshes the page in JavaScript** (window.reload does not exist).

(3) URLs: a **URL gives the location of a resource on the internet**; its parts = **protocol + hostname + directory/filename (all three)** — the HTET version: protocol identifier + www + registered domain name = **all of these**. The **mailto URL sends electronic mail** (asked twice — the e-mail link is written with a href = "mailto: address"). Clicking a **hyperlink opens another document**; **web surfing** = visiting different websites; a site's front page = the **home page**; **e-mail** is the send-and-receive-mail service; **video conferencing** lets users around the globe talk and interact.

(4) Site design and roles: **wireframes produce a rough outline of what the finished website will look like**, and wireframing shows **which content appears on which page AND the navigation between pages (both statements)**. The **webmaster runs the web servers**; **GoLive is a web publishing tool**; **geolocation lets users share their location with trusted websites**; **cookies identify a user who returns to a website**.

(5) Front end vs back end: the front end is what runs in the browser — HTML, CSS, JavaScript; **PL/SQL is NOT a front-end technology** and **PHP is NOT a front-end tool** (server-side); **JavaScript is the front-end/UI choice** in the JS/PHP/Django/SQL list; front-end development tools help build attractive layouts. The 2023 matching: **Front-End Framework → Tailwind CSS, Asset → Font Awesome, Boilerplate → Reset CSS, JavaScript Library → React JS (a-ii, b-iii, c-iv, d-i)**.

(6) Static vs dynamic: a static website serves the same content on every load, is fast, offers no interaction — **the FALSE claim is 'content changes at runtime'** (that is a dynamic site). **DHTML is NOT a programming language** — the **D stands for Dynamic**, and DHTML is the art of combining HTML, JavaScript, DOM and CSS to make pages interactive. **PHP and Apache are BOTH open-source software**.

Exam Points — 11.1

✦ Internet = network of networks (×2), first used for military intelligence; WWW connects/accesses documents; ISP = internet gateway; browser displays pages (MINIX, Acrobat Reader not browsers); search engine locates info (IE is not one); dev tools debug; threads speed image loading; location.reload( ).

✦ URL = resource location; parts = protocol + hostname + directory/filename (all); mailto sends mail (×2); hyperlink opens documents; surfing = visiting sites; home page = front page; wireframes = rough outline + content-and-navigation map; webmaster runs servers; GoLive = publishing tool; geolocation; cookies spot returning users.

✦ PL/SQL and PHP not front-end; JavaScript is; matching a-ii,b-iii,c-iv,d-i (Tailwind/Font Awesome/Reset CSS/React). Static site: content does NOT change at runtime; DHTML = Dynamic, a technique (HTML+JS+DOM+CSS), not a language; PHP + Apache both open source.

11.2 HTML Fundamentals — Structure, Tags and Attributes

HTML: Hyper Text Markup Language — used **to create web pages**. It is a **markup language, not a programming one** (no variables, loops or logic), **derived from SGML** (Standard Generalized Markup Language). **HTML defines the STRUCTURE of a web page** — CSS styles it, JavaScript adds behaviour. **HTML tags must be written within angle brackets < >.**

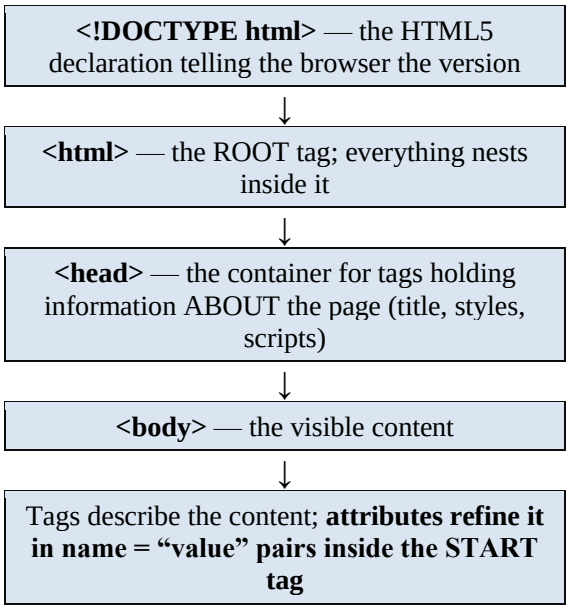


Fig. 11.1 — HTML document structure. A tag contains information about how its content is to be treated.

(1) Attributes: elements can have them; they provide additional information; they come as name = "value" pairs — **the FALSE statement is that attributes are specified in the END tag** (they belong in the start tag). The **class attribute is a global attribute referring to a class in a style sheet** (true statement); **HTML is NOT a server-side processing language** (the false one in that triple); CSS describing display was the third, true.

(2) Comments: written **<!-- like this -->** — option (d) of the 2024 item. **Headings**: HTML supports **6 levels, <h1> (largest, asked three times) down to <h6>**. **
 puts a line break** and is the classic **EMPTY (void) element** — no content, no closing tag (asked twice; the void family: br, hr, img, input, meta, link). The **<hr> horizontal rule** takes SIZE, WIDTH, ALIGN and NOSHADE — **'SHADE' is NOT an attribute**. **RGB(0, 0, 0) displays BLACK** (all zeros; 255,255,255 = white).

Exam Points — 11.2

✦ HTML full form; creates web pages; markup not programming; derived from SGML; tags inside < >; root = <html>; <head> = info about the page (×2); <!DOCTYPE html> declares HTML5; comment = <!-- -->.

✦ Attributes: start tag, name="value" (false claim = end tag); class = global stylesheet attribute; HTML not server-side. Headings h1–h6 (six; h1 largest ×3);
 = line break + void element (×2); <hr> has no SHADE; RGB(0,0,0) = black.

11.3 Links, Images, Semantic Tags and Lists

(1) Hyperlinks: created with the anchor tag — **the HREF attribute holds the destination** (asked as 'parameter to create a hyperlink' and as '<A HREF> jumps from page to page'). **Target values**: _self = same frame, _blank = new tab, _parent = parent frame, **_top = opens the linked document in the window's FULL BODY** (the asked one). **Body-tag link colours**: LINK = unvisited, **VLINK = after the mouse is clicked on it (visited)**, ALINK = active. **Internal linking = linking one part of a document to another part of the SAME document**. An **image map is a special image whose different areas point to different locators on the web**. The **ALT attribute supplies the text displayed when the image cannot be shown** (asked twice — KVS and NVS).

(2) Text and HTML5 semantic tags: **<sup>** = superscript; **<bdo>** changes the current directionality of text; **<cite>** marks titles of works of art, music, movies, books; **<nav>** encapsulates a group of links (HTML5); **<figure>** includes images, graphs and compound objects. The **<meter value="0.8">** item: the text before it displays, the fallback text between the tags does NOT, and a graphical gauge 80 percent filled appears — **all three statements (I, II and III)**. The **name attribute is NOT associated with <EMBED>** (form, select and anchor all take it). An **applet** is a Java program embedded in an HTML document running in a Java-capable browser.

(3) Lists: ** creates an ordered (numbered) list**; the ** unordered list is the bulleted one** — bullets available: **disc, circle and square** (asked twice; no triangle). To switch an ordered list to upper-case letters, use **tag <OL> with the TYPE attribute** (TYPE = A; START only changes the first number).

Exam Points — 11.3

✦ href makes the link (×2); _top = full window body; VLINK = visited-link colour; internal linking = same document; image map = areas → different locators; ALT = fallback text (×2); mailto link = <a href="mailto:…">.

✦ <sup> superscript; <bdo> direction; <cite> titles of works; <nav> link group; <figure> images/graphs; <meter> item = all three statements; <EMBED> has no name attribute; applet = embedded Java program.

✦ ordered; bullets = disc/circle/square (×2); <OL TYPE="A"> for capital letters.

11.4 Tables and Forms

(1) Table structure: <table> encloses all, <tr> is a row, <th> a header cell and **<td> defines the table data cell** (the asked one). The **span attribute defines the number of columns in a column group** (<colgroup>/<col>). **CELLPADDING controls the distance between the data in a cell and the cell's boundaries** (asked twice — KVS 'cellpadding' and NVS 'CELL PADDING'); **CELLSPACING is the gap between cells**. Padding cushions inside; spacing separates.

(2) Forms: a page can hold **any number of forms — there is NO maximum limit** (though forms cannot nest). From JavaScript, **form.submit( ) submits a form** and **event.preventDefault( ) stops it submitting** (cancels the default action in the submit handler).

Exam Points — 11.4

✦ <td> = data cell; span = columns in a colgroup; cellpadding = data-to-boundary distance (×2); cellspacing = between cells; forms per page = no limit; form.submit( ) sends; event.preventDefault( ) cancels.

11.5 CSS — Style Sheets, Selectors and Layout

CSS: Cascading Style Sheets — **a set of style rules that describes a document's display characteristics**. CSS describes how HTML elements are displayed on screen, paper or other media.

Type	Where it lives	Asked wording
Inline	style attribute of one tag	For a single element
Internal	**Inside a <style> element in the <head>**	The asked category (DSSSB PGT 2018)
External	A separate **.css** file	Style definitions saved with the **.css extension**; linked with **<link rel="stylesheet" type="text/css" href="style.css">** (the asked syntax — a <link> tag, not <style> or <a>)

(1) Properties asked: **font-weight specifies the degree of emboldening**; **HSL is the colour specification using hue (0–360°), saturation percent and lightness percent**; **list-style is the shorthand setting multiple list properties at once**; **transitions change property values smoothly over a given duration**.

(2) Tooling and layout: **Sass is one of the most widely used CSS extension languages**; a **CSS framework provides pre-designed UI components and styles** for faster, consistent development; a **CSS pre-processor simplifies and extends the capabilities of CSS**; and **CSS Grid is a TWO-dimensional layout (rows + columns) while Flexbox is ONE-dimensional (a single line)**.

Exam Points — 11.5

✦ CSS = Cascading Style Sheets = display-rule set; internal CSS = <style> in <head>; external = .css file via <link rel="stylesheet"…>; font-weight = boldness; HSL = hue/saturation/lightness; list-style = shorthand; transitions animate smoothly.

✦ Sass = CSS extension language; framework = ready components; pre-processor = simplify + extend CSS; Grid 2-D vs Flexbox 1-D.

11.6 JavaScript — Language, Dialogs and Code Reading

(1) Embedding: JavaScript goes inside the **<script> tag pair** — asked FOUR separate times ('include JavaScript', 'embed scripts', 'suitable tag', 'must enclose those lines'). **Client-side scripting runs code on the USER'S WEB BROWSER** — that is JavaScript's main purpose; a **scripting language is simpler and offers the same protection** (interpreted at run time).

(2) Language basics: variables are created with **var** and **JavaScript is an untyped language — both statements true** (KVS). An identifier may begin with a letter, underscore or dollar — **never a digit, so 10Number is invalid**. **JavaScript arrays can store elements of MULTIPLE data types** (dynamic, not fixed-size). A **callback is a function passed into another function as a parameter**.

Dialog	Behaviour	Asked as
alert( )	Message + OK only	'Displays whatever text you pass' (×2)
confirm( )	Message + **OK AND CANCEL buttons**	The KVS OK/CANCEL question
prompt( )	**Requests text input** from the user	'Send messages requesting text input'

Code-reading corner (all asked in 2025)

(1) let x = 10; if (x > 5) → prints **"Greater"**.

(2) document.write("Hello") then document.write("\nIndia") → the browser collapses the newline to a space: **Hello India**.

(3) The e-mail regex test on "example@domain" → **false** — the pattern demands a dotted extension (like .com) after the domain.

(4) Object.defineProperty with writable: false, then obj.age = 30 → the assignment is silently ignored; console shows **25**.

(5) for...in loops through all enumerable properties of an object; **element.style.backgroundColor = "red" changes CSS dynamically** (the DHTML style method); **window.resizeTo( ) resizes the browser window**.

Exam Points — 11.6

✦ <script> embeds JS (×4); client-side = runs in the user's browser; scripting languages simpler, same protection; var + untyped (both true); 10Number invalid; arrays hold mixed types; callback = function as parameter.

✦ alert = message (×2); confirm = OK + Cancel; prompt = text input. Outputs: Greater; Hello India (newline → space); regex false; writable:false → 25; for...in iterates properties; element.style changes CSS; window.resizeTo( ).

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11.7 The DOM, Events, jQuery and AJAX

DOM: the Document Object Model — a live tree model of the page. **The DOM allows JavaScript to directly modify the structure of the HTML document by manipulating its elements, attributes and content** (the true 2025 statement — it is not read-only and not static).

(1) Object hierarchy: the **WINDOW object is the master container for all content you view in the browser** (the global object); the **DOCUMENT object holds the real content of the page** (root of the DOM tree). Document methods: write, getElementById, getElementsByName, getElementsByTagName — **getElementsByWrite( ) does NOT exist** (the NVS fake).

(2) Events give a web page a DYNAMIC interface — only statement (i) (not static). The 2023 event matching: **onchange → element changed (ii), onmouseover → mouse moves over (iii), onkeydown → key pushed (iv), onload → page finished loading (i)** → a-ii, b-iii, c-iv, d-i. Rasleen's hover-colour link needs the pair **OnMouseOver and OnMouseOut**.

(3) jQuery simplifies JavaScript tasks like HTML document traversal and manipulation; its **slideDown( ) method slides an element down** (reveals it). In AngularJS, **ng-model binds the model data to the view** (two-way binding). **Social-network analysts access Facebook data through Facebook APIs**.

AJAX: Asynchronous JavaScript and XML — the full form asked three times (twice in ASP.NET context, once general). AJAX **sends and retrieves only small amounts of data from the server**, which reduces bandwidth and updates the page without a full reload.

Exam Points — 11.7

- DOM = JS can modify structure/attributes/content; window = master container; document = page content; getElementsByWrite fake; events = dynamic interface (only i); matching a-ii,b-iii,c-iv,d-i; hover pair = OnMouseOver + OnMouseOut.
- jQuery = traversal/manipulation library, slideDown( ); ng-model binds model to view; Facebook APIs; AJAX = Asynchronous JavaScript and XML (×3), sends small data.

11.8 Server-Side Technologies — PHP, ASP.NET, Servlets and CGI

(1) Server vs client: server-side code runs on the web server; the visitor never sees it — **when an ASP.NET file on an IIS server is viewed, the resulting page contains ALL HTML code** (the server sends only its output). **CGI is the standard defining how web servers and application programs communicate**. Web-server services: serving pages, server-side scripting, running gateway programs — **resource sharing is NOT one** (that is an OS job). A **Java servlet receives requests and responds using HTTP — only statement (ii)**; it runs INSIDE a web server, so 'runs without a web server' is false.

(2) PHP: the full form is recursive — **PHP: Hypertext Preprocessor**. A single-line comment starts with // (hash also works). **asort( ) sorts an array ascending WITHOUT affecting the keys** (sort( ) re-indexes; ksort( ) sorts by key); **natcasesort( ) sorts in natural order** ignoring case (item2 before item10). **Passing by reference (&) changes the ORIGINAL value of the argument** when modified inside the function.

(3) ASP / ASP.NET: classic ASP's **default scripting language is VBScript**, whose variables are declared with **Dim**. A server control must carry **runat="server"** — option (d)'s exact syntax. **Data binding** = data from a source bound to controls providing the read/write connection. An **event** = an action or occurrence such as a mouse click, key press or system notification. The **Application object provides global storage accessible to ALL pages of the web application** (Session is per-user). **WPF = Windows Presentation Foundation**; the **Tabular Data Control** is the Microsoft ActiveX control that sorts a data file and produces the information on the page. The **setcookie( ) function stores data in a cookie variable**.

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Exam Points — 11.8

✦ Browser receives ALL HTML from an ASP.NET/IIS page; CGI = web-server-to-program standard; web servers do not do resource sharing; servlet = only (ii), needs a server; PHP recursive full form; // comments; asort keeps keys; natcasesort natural order; & reference edits the original.

✦ ASP default = VBScript, Dim declares; runat="server"; data binding = source-to-controls; event = click/key/notification; Application object = global state for all pages; WPF = Windows Presentation Foundation; Tabular Data Control; setcookie( ) stores cookie data.

11.9 XML, DTD and Schema

(1) XML's common use: to represent and exchange data between different systems and platforms — self-describing, platform-independent text. Its key components: **tags, attributes and comments**. The prolog opens with the declaration **<?xml version="1.0"?>** — asked twice (KVS 2013 and 2017), both keys choosing the question-mark form. The single top-level element containing all nested sub-elements = the **ROOT element** (an XML document has exactly one). In XML, the **id attribute uniquely identifies an element**.

(2) Naming rules: an element name **must NOT start with the letters 'xml'**, cannot start with a number, cannot contain spaces — and the 2023 statement 'tag names can begin with a numeric value' is **FALSE**.

(3) DTD = Document Type Definition (full form asked). A **DTD defines the structure — the elements and attributes a document may contain** (asked twice: 'defines a document type', 'defines structure and data types'). DTDs are **the declarations that define a document type for SGML** (true), and **XML Schemas support datatypes while DTDs do NOT** (true) — the 2023 triple keyed i-True, ii-True, iii-False. Element declaration syntax: **<!ELEMENT element-name category>**; an external DTD is referenced as **<!DOCTYPE root-element SYSTEM "file-name">**.

(4) Parsers: an **XML processor reads XML documents and provides access to their content and structure**; the **function of an XML parser is to convert XML content into XML DOM content**. The **DOM parser is the object containing all the document's information composed like a TREE structure** (whole tree in memory, random access); SAX streams events; JDOM is real — **'Shell' is NOT an XML parser**.

Exam Points — 11.9

✦ XML exchanges data between systems; components = tags + attributes + comments; prolog = <?xml version="1.0"?> (×2); root element = single top-level (×2); id = unique identifier; names: never start with xml or a digit, no spaces.

✦ DTD full form; defines document structure (×2); DTD-for-SGML true + Schema-has-datatypes true (i-T, ii-T, iii-F); <!ELEMENT name category>; <!DOCTYPE root-element SYSTEM "file">; processor reads + gives access; parser → XML DOM content; DOM = tree in memory; Shell not a parser.

Quick Revision — Chapter 11 (30 Points)

1. Internet = network of networks (first use: military); WWW = connect and access documents; browser displays pages, search engine finds them (IE = browser; MINIX, Acrobat Reader not browsers).
2. URL = location of a resource: protocol + hostname + directory/filename; mailto URL sends e-mail; hyperlink opens another document; home page = front page.
3. Wireframes = rough outline of the finished site (content per page + navigation); webmaster runs servers; GoLive = web publishing tool; cookies identify returning visitors; geolocation shares location.
4. Front end = HTML + CSS + JavaScript (PL/SQL and PHP are back-end); matching: framework–Tailwind, asset–Font Awesome, boilerplate–Reset CSS, JS library–React.
5. Static site never changes at runtime; DHTML = Dynamic HTML + JS + DOM + CSS, a technique not a language; PHP and Apache both open source.

6. HTML = Hyper Text Markup Language, creates web pages, derived from SGML; markup not programming; tags in < >; root = <html>; <head> holds page information; <!DOCTYPE html> = HTML5.
7. Comments <!-- -->; attributes = name="value" pairs in the START tag; class = global stylesheet attribute; headings h1 (largest) to h6;
 = line break, void element; <hr> has NOSHADE not SHADE; RGB(0,0,0) = black.
8. href builds links; _top = full window; VLINK = visited colour; internal link = same document; image map = areas to different locators; ALT = text when image fails.
9. HTML5: <bdo> direction, <cite> work titles, <nav> link group, <figure> images/graphs, <meter> gauge (all three statements); <sup> superscript; <EMBED> takes no name; applet = embedded Java.
10. numbered (TYPE=A for capitals); bullets = disc, circle, square; list-style = CSS shorthand.
11. <td> = data cell; span counts colgroup columns; cellpadding inside the cell, cellspacing between cells; unlimited forms per page; form.submit( ) and event.preventDefault( ).
12. CSS = Cascading Style Sheets = display rules; inline / internal (<style> in head) / external (.css via <link rel="stylesheet">).
13. font-weight = boldness; HSL = hue 0–360 + saturation + lightness; transitions animate smoothly; Sass = CSS extension; framework = ready components; pre-processor extends CSS; Grid 2-D, Flexbox 1-D.
14. JavaScript lives in <script>; client-side = runs in the user's browser; var declares, untyped language; identifiers never start with a digit; arrays hold mixed types; callback = function passed as parameter.
15. alert = message; confirm = OK + Cancel; prompt = text input.
16. Code outputs: x=10 → Greater; document.write pair → Hello India; strict e-mail regex on example@domain → false; writable:false keeps 25; for...in walks properties.
17. element.style.backgroundColor changes CSS live; window.resizeTo( ) resizes; location.reload( ) refreshes.
18. DOM = live tree; JS modifies structure, attributes, content; window = master container, document = page content; getElementsByWrite does not exist.
19. Events make the page dynamic; onchange/onmouseover/onkeydown/onload matching a-ii,b-iii,c-iv,d-i; hover pair OnMouseOver + OnMouseOut.
20. jQuery simplifies DOM traversal (slideDown slides open); ng-model binds model to view; Facebook data via APIs.
21. AJAX = Asynchronous JavaScript and XML; exchanges small amounts of data in the background — no full reload.
22. Server-side output only: an ASP.NET page reaches the browser as pure HTML; CGI = server-to-program standard; web servers do not share resources; servlets run inside a server and speak HTTP.
23. PHP: Hypertext Preprocessor (recursive); // single-line comment; asort keeps keys; natcasesort natural order; & passes by reference (edits the original).
24. ASP: VBScript default, Dim declares; ASP.NET controls need runat="server"; data binding ties data to controls; event = click/key/system action.
25. Application object = global state for every page; Session = one user; setcookie( ) stores cookie data; WPF = Windows Presentation Foundation; Tabular Data Control extracts file data.
26. XML exchanges data between systems; built of tags, attributes, comments; one root element; id unique; prolog <?xml version="1.0"?>.
27. XML names: never start with xml or a digit, no spaces.
28. DTD = Document Type Definition, defines the permitted structure; DTDs came from SGML; XML Schema adds datatypes that DTD lacks.
29. <!ELEMENT name category> declares; <!DOCTYPE root-element SYSTEM "file"> attaches an external DTD.
30. XML processor/parser reads the document into XML DOM content; DOM parser = whole tree in memory; SAX streams; JDOM real, Shell fake.

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